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M.C.A. (Management)

MT - 21 : OPTIMIZATION TECHNIQUES

(OTE 552 MJ) (2024 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates :

- 1) All questions are compulsory.
- 2) Use of statistical table and non-programmable calculator is allowed.
- 3) Figures to right indicate full marks.

Q1) A) Solve the following (Any One) : [8]

- a) Solve with two phase simplex method

$$\text{Maximize } Z = 3X_1 + 4X_2$$

$$\text{Subject to } 2X_1 + 3X_2 \leq 200$$

$$5X_1 + 4X_2 \geq 100$$

$$-8X_1 - 4X_2 \leq -80$$

$$\text{Such that, } X_1, X_2 \geq 0$$

- b) A company has three plants W, X and Y and three warehouses A, B, C. The supplier is transported from the plants to the warehouses which are located at varying distances from the plants. Due to this, the transportation costs from plants to warehouses vary from Rs.8 to Rs.24 per unit. The company wishes to minimize the transportation costs. The costs (in Rs.) from the plants to the warehouses are shown in the form of a matrix. Determine the optimum shipping schedule. Use LCM for initial solution.

Plant	Warehouse			Supply
	A	B	C	
W	12	8	18	400
X	20	10	16	350
Y	24	14	12	300
Demand	500	200	300	

P.T.O.

B) Explain the following (Any One) : [2]

- a) Infeasible Solution
- b) Slack and Surplus Variable

Q2) A) Solve the following (Any One) : [5]

- a) There are 7 jobs each of which must go through three machines M1, M2 and M3. The corresponding processing time on the machines in hours are as follows

Job	J1	J2	J3	J4	J5	J6	J7
M1	3	8	7	4	9	8	7
M2	4	3	2	5	1	4	3
M3	6	7	5	11	5	6	12

- i) What is the optimal job sequence that will minimize the total elapsed time?
 - ii) What is the idle time for each machine (M1, M2 and M3)?
- b) There are seven jobs, each of which has to go through machines A and B in the order BA. Processing time in hours as follows:

Determine an optimum sequence of these jobs that will minimize the total elapsed time

(T). Also find T and calculate idle time for machine A and B

Job	1	2	3	4	5	6	7
Machine A	3	12	15	6	10	11	9
Machine B	8	10	10	6	12	1	3

B) Solve the following (Any One) : [5]

- a) The person works in the internet café works for 8 hours a day. The rate of arrival of customers at an internet café follows Poisson distribution with average of fifteen minutes between one customer and the next. The duration of service is assumed to follow exponential distribution with mean of twelve minutes.
 - i) What is the probability that person arriving at the café will have to wait?
 - ii) Find the total time for which the person will be idle

- b) In a bank with a single cashier, customers arrive on average every 20 minutes and the cashier takes 10 minutes to attend to each customer.
- What is the average number of customers waiting for the cashier?
 - What is the average time spent by each customer in the bank?

Q3) A) Solve the following (Any One) : [5]

- a) Three companies introduced new razor blades in the market at the same time, each initially holding an equal market share. After the first year, some changes occurred, which are shown by the following transition matrix:

	A	B	C
A	0.9	0.03	0.07
B	0.2	0.6	0.2
C	0.1	0.2	0.7

Assume no changes in customer buying habits occur. What will be the market share of each company at the end of the first and second year?

- b) Two companies, A, B compete against each other. The transition matrix T for people switching each month among them is given by the following transition matrix.

		NEXT	
		A	B
INITIAL	A	65%	35%
	B	45%	55%

Find the long term distribution

B) Solve the following (Any One) : [5]

- a) The daily demand for Coca-Cola bottles, based on past experience, is given as follows:

Daily Demand	10	20	30	40	50
Probability	0.15	0.10	0.40	0.20	0.15

A manager decides to keep 25 Coca-Cola bottles in stock. Using the following random numbers: 69, 01, 08, 74, 82, 20, 72, 14, 75, 12, 25 estimate the average balance stock.

- b) The Lajwaab Bakery shop keeps stock of a popular brand of cake previous experience indicates the daily demand as given below

Daily Demand	0	15	25	35	45	50
Probability	0.01	0.10	0.30	0.45	0.12	0.02

Consider the following sequence of random numbers 21, 28, 57, 54, 60, 39, 74, 91, 75, 30. Using this sequence simulate the demand for next 10 days. Find out the stock situation if the owner of the bakery shop decides to make 30 cakes every day.

Q4) A) Solve the following (Any One) :

[8]

- a) Consider a CPM network with the following project data:

Activity	Duration (days)	Predecessor(s)
A	4	None
B	3	None
C	6	A
D	2	B
E	5	C,D
F	3	D
G	7	A,B
H	5	E,F

- i) Draw a network diagram. Perform Forward Pass and Backward Pass calculations to determine the earliest start (ES), earliest finish (EF), latest start (LS) and latest finish (LF) for each activity.
- ii) Calculate the total float and free float for each activity. Identify the activities that are on the critical path.
- b) A project consists of 9 activities whose time estimates in weeks & other characteristic are given below

Activity	Preceding activity	most likely time (Week)	Pessimistic time (Week)	Optimistic time (Week)
A	--	4	6	2
B	---	6	6	6
C	--	12	24	6
D	A	5	8	2
E	A	14	23	11
F	B,D	10	12	8
G	B,D	6	9	3
H	C,F	15	27	9
I	E	10	16	4

- i) Draw the PERT network & find the critical path (Perform Forward Pass and Backward Pass calculations)
- ii) find the probability that the project is completed 1 week before the expected time

Q4) B) Explain the following (Any One) : [2]

- a) Difference between PERT and CPM
- b) Dummy Activity

Q5) A) Solve the following (Any One) : [5]

- a) The following table gives profit matrix for different strategy and actions. Find the optimal strategy by using EMV Criterion

States of Nature	Probability	Actions		
		A1	A2	A3
E1	0.15	40	52	45
E2	0.25	70	28	40
E3	0.45	30	70	-50
E4	0.15	30	-50	-70

- b) For a given Cost matrix, suggest optimum strategy using-

		Events			
		N1	N2	N3	N4
Strategies	S1	1000	1500	750	0
	S2	250	2000	3750	3000
	S3	-500	1250	3000	4750
	S4	-1250	500	2250	4000

- i) Hurwicz Criterion (Alpha = 0.7)
- ii) Savage Criterion

Q5) B) Solve the following (Any One) : [5]

- a) Define a zero-sum game and explain the concept of dominance in game theory. Solve the following 2 x 2 zero-sum game using dominance:

	Strategy 1	Strategy 2
Player 1	6	4
Player 2	5	3

Identify the dominated strategy (if any) for both players
Simplify the game using the dominance rule and find the optimal strategies for both players.

b) Solve the following to find the value of the game

		PLAYER B				
		B1	B2	B3	B4	B5
PLAYER A	A1	2	4	3	8	4
	A2	5	6	3	7	8
	A3	6	4	9	8	7
	A4	4	2	8	4	3

